



March 15, 2025

Office of Science and Technology Policy (OSTP)
Executive Office of the President
1650 Pennsylvania Avenue NW
Washington, DC 20504

Subject: Response to Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Dear OSTP Team,

VEIR appreciates the opportunity to provide input on the development of the AI Action Plan. The growth of AI-driven data centers represents a significant means for America to maintain its global leadership in AI. However, the rapid increase in AI data center electricity use necessitates that the nation's energy infrastructure, particularly our grid, keeps pace. We must expand our electricity systems rapidly while also ensuring that electricity remains affordable. The electricity load growth and associated infrastructure investments being driven by AI present an opportunity to double down on Department of Energy (DOE) Secretary Wright's objectives of "strengthening grid reliability and security" and prioritizing "true technological breakthroughs," while keeping electricity affordable for Americans. Furthermore, as more advanced AI chips are deployed, the power density requirements for data centers are increasing beyond the capabilities of traditional power delivery equipment. This ultimately affects the ability of data centers to maximize computing performance and infrastructure investment.

VEIR's ultra-high capacity, superconducting transmission can help the power grid scale quickly to meet upcoming demand by offering five to ten times the capacity within the same space compared to conventional technologies used today. Additionally, VEIR's indoor power delivery solutions provide twice the power over ten times longer distances, using one tenth of the space and weight that would be required by conventional aluminum or copper busbars and busways. This offers data centers new ways to configure buildings to maximize computing capability. Advanced, high-density, power delivery solutions such as VEIR's are an essential component for maintaining U.S. leadership in AI innovation, economic growth, and national security.

Background on VEIR

VEIR is pioneering high-density power delivery solutions that use superconductors and advanced cooling systems to enable the rapid scaling of new or existing power delivery systems, while providing high reliability and flexibility in design. VEIR's high-density power delivery solutions take up less space than traditional solutions, allowing users to deploy new systems

faster or significantly expand capacity within existing spaces. VEIR's systems also actively manage heat, providing consistent amounts of power regardless of ambient temperature. These advantages make VEIR's technology particularly well-suited to meet the needs of AI data centers, which require reliable, high-capacity power solutions from grid to chip to sustain operations and growth.

VEIR's technology scales across voltage classes, from 48 V to 275 kV, and VEIR offers both high-voltage transmission lines to expand capacity in the grid as well as power delivery systems within data centers and on data center campuses to increase computing capability. VEIR's superconducting technology can deploy in multiple AI-related applications, including:

- Connecting new AI data centers and campuses faster or expanding the interconnection of existing facilities.
- Optimizing power connections between data center buildings and on-site generation assets for higher reliability and resiliency.
- Rapidly enhancing the resilience of regional grids serving AI hubs by enabling the significant capacity expansion of existing lines or quick deployment of new lines.
- Optimizing AI data center layouts and performance by enabling high-power computing clusters and flexible configurations.¹

Prioritizing High Power Density Solutions to Accelerate AI Data Center Deployment

The ability to connect AI data centers to a robust grid in a timely manner is essential to U.S. leadership in AI, and the ability to scale AI data centers while maximizing compute power will only make the U.S. more competitive. However, the current process for expanding transmission capacity is often slow and often out of sync with the rapid development cycles of new AI infrastructure. Furthermore, trialing newer indoor technologies requires the opportunity to engage in ongoing demonstrations and projects.

Priority Actions for the AI Action Plan

To accelerate the deployment of high-power density superconducting solutions, and ensure the nation's AI growth, VEIR recommends the following actions be included in the AI Action Plan:

1. Federal Support to Unleash American Energy Innovation

- Recognize superconductors as a strategic priority in national energy and AI infrastructure planning.
- Provide funding for research, development, and deployment of modern superconducting transmission technologies on the grid.

¹ VEIR indoor power delivery solutions create greater design flexibility by being able to span longer distances and take up less space than conventional busbars and cables. This also enables more servers to be spaced more closely, reducing latency and improving aggregate AI server performance.

- Provide funding for research, development, and deployment of modern superconducting technologies that enable higher power density and higher performance of AI data centers.
- Facilitate collaboration between federal agencies, national labs, utilities, and AI data center developers to advance superconducting solutions.

2. Streamlining Regulatory Approvals for High Power Density Solutions

- Require all federal agencies to adopt DOE's National Environmental Policy Act (NEPA) categorical exclusion for reconductoring and rebuilding transmission lines.
- Establish a specific categorical exclusion for superconducting transmission projects to expedite permitting. Superconductors can leverage existing rights-of-way to increase capacity or offer significant capacity at voltages often excluded by state review requirements.

Conclusion

Superconducting technology represents a transformative solution for addressing the energy demands of AI data centers on the grid and within their walls. By explicitly supporting high-power-density transmission solutions in the AI Action Plan, OSTP can help drive innovation, strengthen grid reliability, and position the U.S. as a leader in AI-enabled economic growth. VEIR welcomes the opportunity to serve as a resource to OSTP and looks forward to further collaboration on advancing AI and energy infrastructure policy.

Sincerely,

Tim Heidel
CEO
VEIR



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